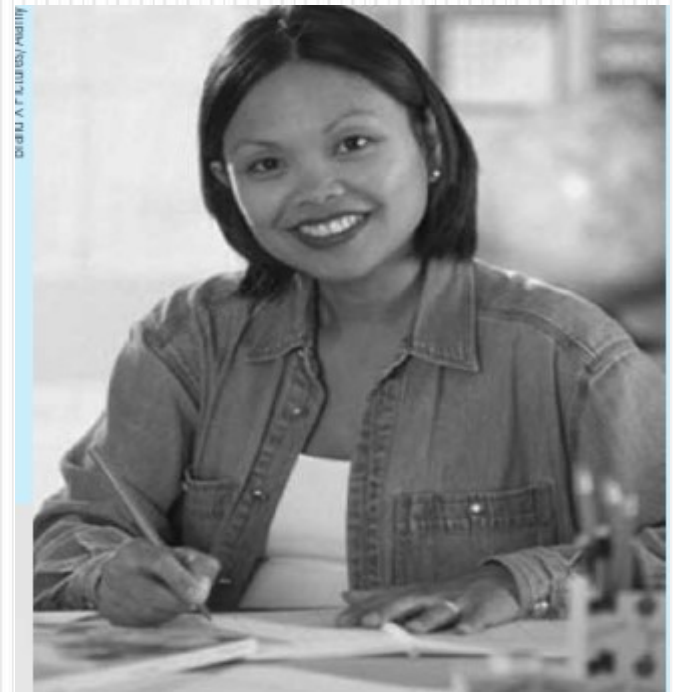


Chapter 9

Adult Nutrition



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Objectives

- After completing this chapter, you will be able to:
- Describe the physiological & hormonal changes during Adulthood
- Discuss the nutrient recommendations for Adults.
- Nutrition transition
-

Definition of Adulthood in the Life Cycle

- Adulthood is a period of optimum mental functioning when the individual's intellectual and social performance are at their peak
- Divisions of ages in adulthood used in are:
 - Early adulthood—ages 20-39 yrs
 - Middle adulthood—ages 40 to 64 yrs
 - Old age—age 65+

Importance of Nutrition

- Foods have many roles:
 - A source of fuel
 - For enjoyment & comfort
 - A symbol of traditions, rituals, & celebrations
- How do food & nutrition enhance life?
 - Nutrition & exercise are among the main lifestyle factors that reduce risk of premature death

Physiological Changes During Adulthood

- For the most part, individuals have stopped growing by the time they reach their twenties.
- Continue to develop bone density until age of 30.
- Muscular strength peaks around 25 to 30 years of age.
- By middle adulthood, decline in size and mass of muscles and an increase in body fat.
- Hearing loss begins as early as age 25 and vision changes often become noticeable by age of 40.

Physiological Changes of Adulthood

- **Nutritional emphasis:**
 - maintaining physical status (muscle strength)
 - avoiding excess fat (especially around the midriff)
- Adequate nutrition is needed to replace body cells



Hormonal Changes



Women

- Hormonal changes linked to **menopause**

Concerned about not gaining weight & looking good

Men

- Hormones linked to **climacteric** change

Concerned about strength, energy, & weight management

Physiological Changes in Males

Climacteric



- Gradual decline in testosterone begins at age 40 to 50
- Dietary factors linked to changes:
 - Underweight → ↓ sperm
 - Malnutrition → ↓ libido
 - Alcohol → defective sperm
- Body weight rises beginning ~age 40

Physiological Changes in Females

Menopause



- With cessation of menstruation, iron needs ↓ from 18 to 8 mg/day
- Loss of estrogen leads to:
 - Atrophy of urinary tract & vagina
 - Increased abdominal fat
 - Risk of osteoporosis & heart disease

Body Composition Changes

- Associated with a **+ve** energy balance with an increase in **weight and adiposity** and a **↓** in **muscle mass**.
- Redistribution of fat occurs with gains in **central and intra-abdominal space**.
- This redistribution of body fat is associated with increased risk for hypertension, insulin resistance, diabetes, stroke, gallbladder disease, and coronary artery disease.

Maintaining a Healthy Body

Energy for Weight Management

- Caloric expenditure declines ~20% during adulthood
 - Due to ↓ in metabolic rate & ↓ activity level
- Energy expenditure remains constant IF fat-free body mass stays constant

Alcohol: Food, Drug and Nutrient

- CSA of Ethiopia, involving age group 15–49 year:
- 48 % of women and 53 % men of drinkers reported consumption of alcohol six or more days in the past month
- Highest rate is among those ages 25-44

Nutrient Recommendations

- Daily nutrient intakes for older adults are similar to those for younger adults.
- The exception is energy; the need decreases with age.
- Recommendations for acceptable macro nutrient distribution ranges for total calorie intake are:
 - Fat 20–35% of calories
 - Carbohydrate 45–65% of calories
 - Protein 10–25% of calories.

Energy

- Energy requirements ↓ as one ages for three main reasons.
 - Change in composition to relatively more lipid & less muscular tissue.
 - The reduction in active tissue
 - Decrease in the basal metabolic rate
- In addition as people grow older, there will be a decline in physical activity and they tend to eat less.

Protein

- The recommended level of protein intake is 0.8 g/kg body weight.
- Foods of high protein density, such as eggs and meat, should be included in the diet of adults.

Fiber

- Fiber, especially soluble or viscous fiber:
 - reduces absorption of **cholesterol from the gut** and has positive effects on blood lipids, and weight management.
- Average daily recommendations are 38 g for men and 25 g for women, or 14 g per 1000 calories

Water and Other Fluids

- Adults can survive 3-4 days without water
- Fluid intake recommendations are 1 mL/kcal consumed
 - At least 1500 mL if < 1500 kcal consumed
 - In the average person, **water in foods** accounts for **~19%** of fluid with **~81%** from beverages

Water Intake Recommendations

- Adult adequate intake (AI) level for water is total water intake (from fluids and food).
- **Total Water AI for Adults**
 - Men 3.7 liters
 - Women 2.7 liters
- So, on average, men and women need to drink approximately 12 cups and 9 cups each day, respectively

Vitamins

Vitamin A



- We need vitamin A for healthy skin and mucus membranes, our immune system, and good eye health and vision.
- Antioxidant
- DRI 700-900mcg/d
- Sources of beta-carotene (precursor of vitamin A), include sweet potatoes, carrots, pumpkin, broccoli, and green leafy vegetables.

Vitamin D



Exposure of skin to sunlight is an important

- Food sources of vitamin D include fish, milk, egg yolk
- The AI for vitamin D is **5 mcg** (μg)/day up to age of 50, and then it doubles to **10 mcg**.
- Vitamin D deficiency is a highly prevalent condition in polar areas



Vitamin E

- Research suggests that vitamin E protect **prostate & esophageal cancer, heart disease**
- The adult RDA is 15 mg.
- Food sources of vitamin E include vegetables oils (e.g. corn, safflower, olive oils), sunflower seeds, wheat germ, salad dressings made with oil, avocado and leafy greens.



Choline

- Choline has a role in preventing fatty liver.
- AI 425-550 mg/d
- It has many functions, including **lipid transport**, and along **with folate and vitamin B12** is involved in converting homocysteine to methionine .
- Liver, eggs, beef, broccoli, milk, and peanuts are food sources of choline



Calcium



- Calcium, vitamin D and magnesium help develop and maintain bone density
- The recommended dietary calcium intake for elderly individuals is 5mcg
- food source kale, milk products, bean and lentils



Magnesium



- Poor diet or prolonged illness can increase the risk of magnesium deficiency, with adverse effects on bone strength, heart health
- DRI 310-420 mg/d
- Good sources are sunflower seeds, beans, milk, whole grains, and bananas.

Potassium

- Is important for electrolyte balance and muscle contraction & reduce blood pressure.
- DRI 4.7g/d
- Good sources include wheat bran, wheat germ, potatoes, bananas, and orange juice.



Iron



- Iron deficiency causes of nutritional anemia.
- For women, iron needs drop from 18 to 8 mg per day at menopause
- High-iron foods include red meats, fortified cereals, and dried beans.
- Enriched breads and most fruits and vegetables also contribute iron.
- .



Nutrition and Health Promotion

General principle for healthful nutrition habits:

Variety & moderation in choosing a diet that will achieve & maintain a healthy body weight when combined with activity

Physical Activity Recommendations

- Lack of physical activity is associated with higher risk of death or disability



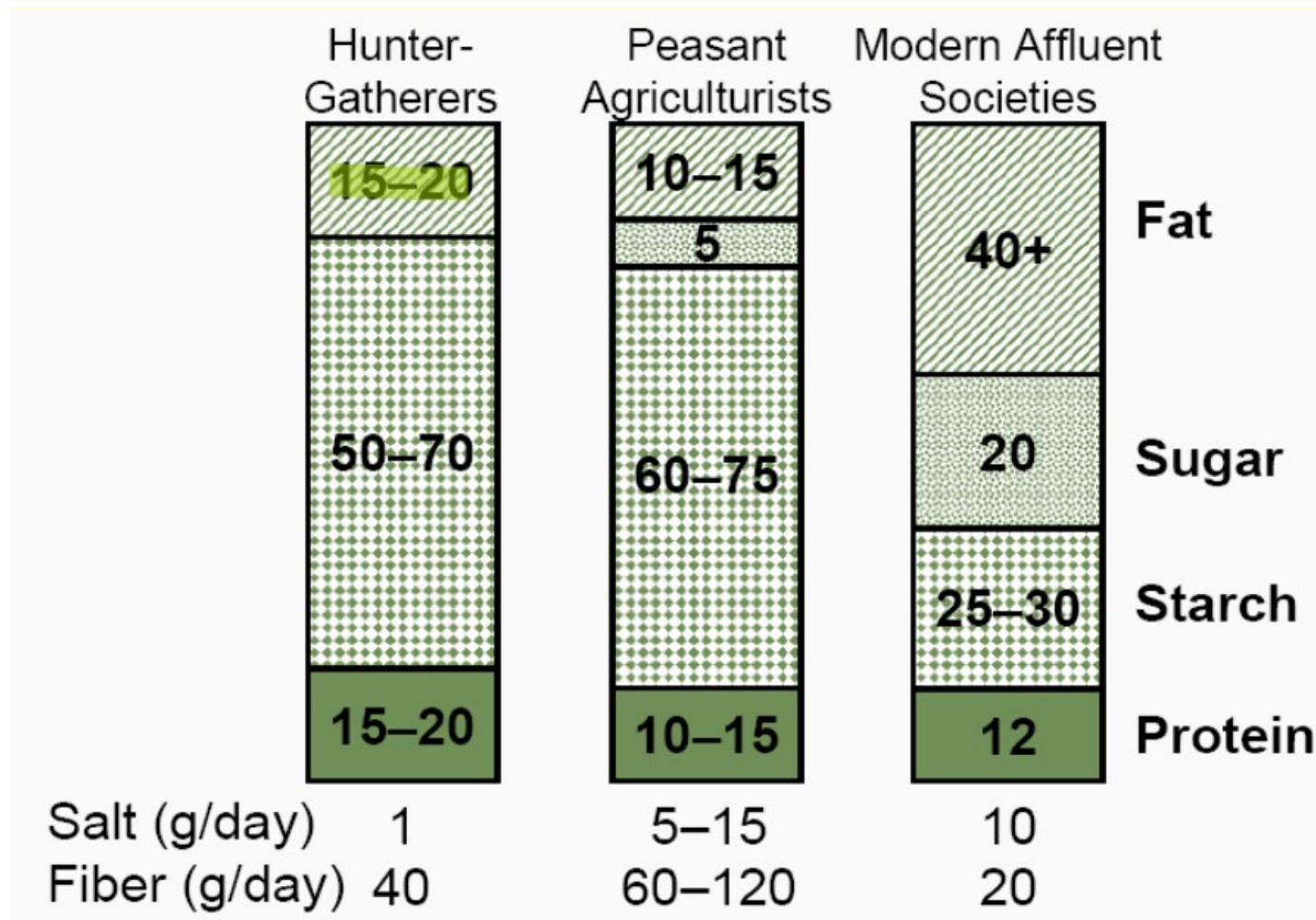
- At least 30 minutes/day of physical activity
- Resistance training 2 or 3 times a week for 30 minutes/day



The Nutrition Transition

- The demographic changes are accompanied by urbanisation, economic growth, technological innovations, mechanisation coupled with reduced physical activity and higher leisure time of sedentary nature.

Evolution of Human Diet



The Nutrition Transition

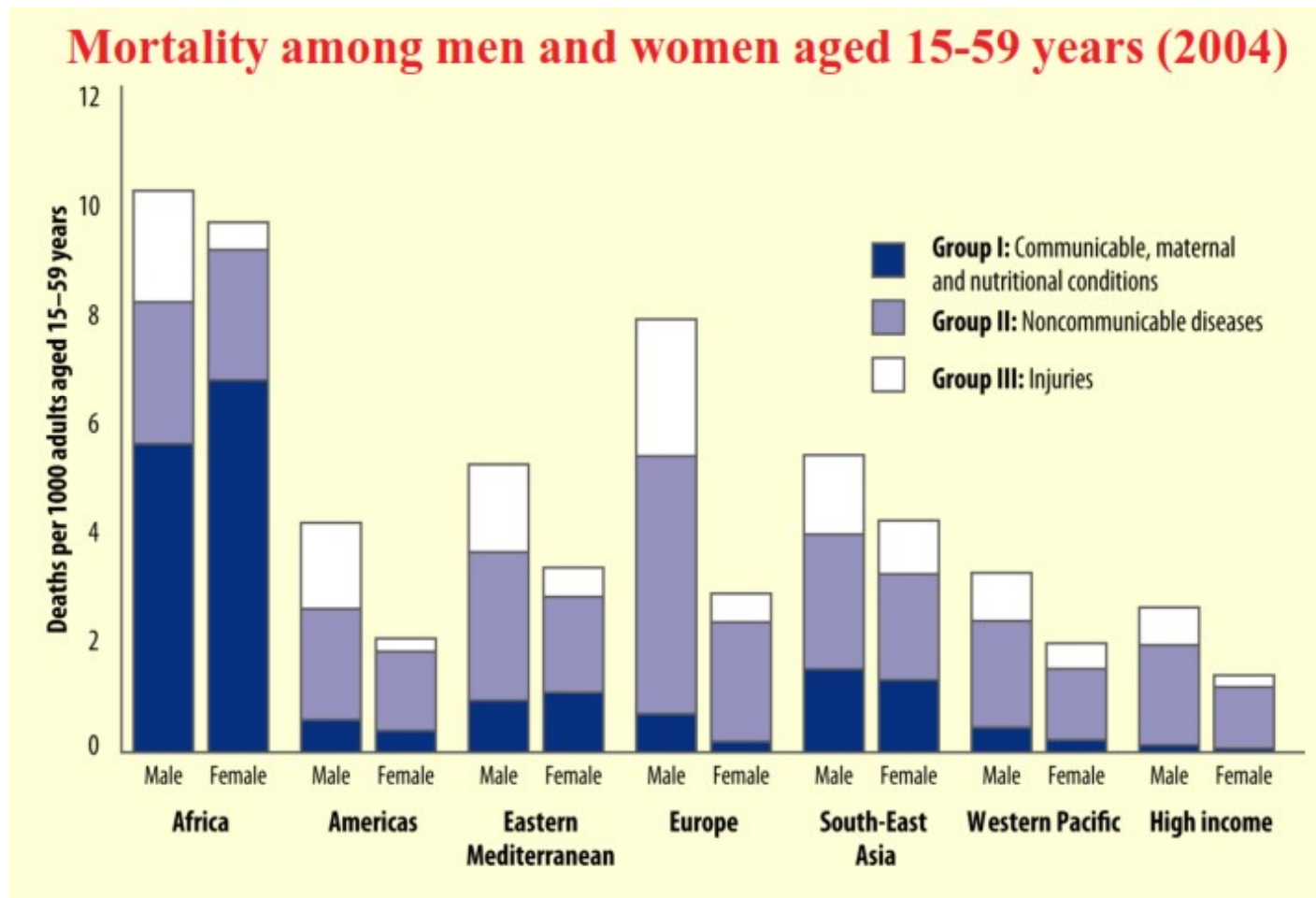
- ✚ The **starchy foods of low calorie value with high fibre and micronutrient diets** are being replaced with **calorie dense, fatty, salty, low fibre and more animal foods**, which are loaded with **saturated fats**.
- ✚ The children in urban areas are becoming obese due to both
 - ✚ lack of physical activity and
 - ✚ easy availability and access of *junk* food,

Disease states associated with
unhealthy nutrition and
lifestyle

Introduction

- Life-style conditions, the leading causes of death in adults, are :
 - Cancer
 - Cardiovascular diseases
 - Overweight & obesity
 - Diabetes
 - Metabolic syndrome

Noncommunicable Diseases



Cancer

- Definition of cancer
 - a group of diseases in which abnormal cellular growth affects specific organ systems
- Tissues that are affected:
 - Connective—sarcomas & leukemia
 - Muscular—myosarcomas
 - Nervous—gliomas
 - Epithelial—the main→ skin, GI, urogenital, secretory, & respiratory

Prevalence/Etiology of Cancer

- 4 most common cancers for adults are
 - prostate, breast, colorectal, & lung
- Blacks—highest rate of new cancers

Risk Factors of Cancer

- Smoking—linked to 30% of cancers
- Nutrition-related risks
 - Obesity & insulin resistance
 - ↑ alcohol consumption
 - ↓ intakes of fruits, vegetables, & calcium
- Diet/food related risks
 - ↑ fat
 - Charred foods (grilled meats)
 - *Helicobacter pylori*

Nutrition Intervention for Cancer

- To minimize nutritional risk:
 - Maintain or reach healthy weight
 - Eat 3-4 servings of fruits & vegetables
 - Eat a low-fat diet
 - Alcohol in moderation, if at all
 - Exercise
- Health care professionals provide guidance on diet/food-related side-effects of treatment

Cardiovascular Disease: Coronary Heart Disease

- Diseases related to the heart and blood vessels
- Cardiovascular disease (CVD) consists of coronary heart disease (CHD) , stroke, & other disorders of heart, blood vessels & circulation
- Hyperlipidemia (high blood cholesterol) and hypertension are important factors

Definition of CHD

- CHD and stroke are characterized by atherosclerosis, hardening of arteries, due to plaque buildup
- Arteries narrow increasing risk of:
 - Myocardial infarction → shut off blood in heart
 - Stroke → blood shut off to cerebral artery of the brain

Etiology of CHD

- Begins when substances like cholesterol, fatty acids & calcium accumulate over injured arterial wall cells
- ↑ levels of homocysteine, inflammatory diseases, abnormal blood clotting factors, etc.

Effects of CHD

- Buildup of plaque & lesion leave less room for blood flow
- Results may be:
 - Decreased blood flow to heart
 - Reduced energy
 - Decline in organ function
 - Shortness of breath
 - Chest pain

Effects of Obesity

↑ body weight = ↑ mortality

Overweight & obesity associated with:

- Hypertension & stroke
- Dyslipidemia & coronary heart disease
- Type 2 diabetes
- Gallbladder disease
- Osteoarthritis
- Sleep apnea
- Back problems
- And many more...

Diabetes Mellitus

- Definition: fasting blood glucose levels ≥ 126 mg/dL
- Type 1
 - Pancreatic beta cells destroyed
 - Daily insulin injections required
- Type 2
 - Insulin resistance related to obesity
 - Most common type
 - ~200 million world wide

Etiology of Diabetes

- Type 2 caused by insulin resistance + insulin deficiency
 - Insulin:
 - Hormone produce by pancreatic beta-cells
 - Facilitates passage of glucose into cells
- Adverse effects of high blood glucose:
 - ↑ blood triglycerides
 - ↑ blood pressure
 - Hardening of arteries

Insulin Resistance

- What is insulin resistance?
 - Condition in which cells “resist” the action of insulin in facilitating the passage of glucose into cells
- Contributors to insulin resistance:
 - Obesity
 - ↓ physical activity
 - Genetic predisposition
 - Abdominal obesity

Effects of Diabetes

Short Term

- Blurred vision
- ↑ urination
- ↑ infections
- ↓ wound healing
- Hunger & thirst

Long Term

- Heart disease
- Hypertension
- Stroke
- Blindness
- Kidney failure
- Poor circulation
- Loss of limbs

Risk Factors

- Main risk factors for type 2
 - Obesity & central obesity
 - Physical inactivity
 - ↓ whole grains & fiber
 - Genetics

↓weight + ↑exercise = ↓ risk

Nutrition Interventions for Type 2 Diabetes

- Diet should consist of:
 - Whole grains & other fibrous foods
 - Unsaturated fats
 - Regular meals & snacks
 - CHO counting

Metabolic Syndrome

- A cluster of altered metabolic conditions that come together in a single individual.
- Associated with ***hyperinsulinemia***
- The metabolic conditions include
 - abdominal obesity,
 - high blood pressure,
 - elevated fasting glucose, dyslipidemia with elevated LDL ("bad") cholesterol, low HDL ("good") cholesterol, and
 - elevated triglycerides

Factors that contribute to the development of MS

- Genetics and the environment
- Lifestyle issues such as low activity level, sedentary lifestyle, and progressive weight
- A pro-inflammatory, atherogenic diet high in total and saturated fat and low in whole grains, vegetables, and fruits
- Metabolic abnormalities that ↑ risk of type 2 diabetes including
 - ↑ central body fat
 - ↑ blood insulin, fasting glucose, TG, BP
 - ↓ HDL

Screening and Assessment

- Waist circumference
- A fasting lipid profile providing LDL cholesterol and HDL cholesterol and triglycerides levels, and
- Fasting blood glucose, along with BP

Nutrition Interventions

- The first line therapy is directed to the risk factors:
 - for dyslipidemia, achieve an optimal lipid profile;
 - for hypertension, normalize blood pressure
 - for elevated glucose, reduce fasting blood glucose and increase insulin sensitivity.
- The preferred treatment is:
 - lifestyle change
 - adopting healthy eating, and
 - reducing weight